

KUMASI TECHNICAL UNIVERSITY COMPUTER SCIENCE DEPARTMENT

GUIDELINES FOR WRITING UNDERGRADUATE PROJECT REPORT

The Document provides the guidelines for writing project reports conducted by undergraduate students of the Computer Science Department, Kumasi Technical University.

TABLE OF CONTENTS

AIMS OF THE PROJECT	3
INTENDED LEARNING OUTCOMES	3
<i>Assessable outcomes</i>	<i>3</i>
<i>Additional outcomes.....</i>	<i>3</i>
PROJECT SUPERVISION ACQUISITION.....	3
BRIEF DESCRIPTION OF TEACHING AND LEARNING METHODS	3
ACADEMIC MISCONDUCT	4
<i>Copying</i>	<i>4</i>
<i>Cheating.....</i>	<i>4</i>
FORMATTING YOUR PROJECT.....	4
STRUCTURE OF THE REPORT.....	5
PROJECT OUTLINE / CONTENTS OF CHAPTERS.....	10
Chapter One: Introduction	10
Chapter Two: Literature Review	10
Chapter Three: Methodology and System Design	10
Chapter Four: System Implementation, Testing and Results.....	11
Chapter Five: Discussion, Conclusion and Recommendations	11
Final Submission Requirements	12
ASSESSMENT.....	12
WRITING YOUR PROJECT PROPOSAL	12
Title Page Format.....	12
Content of the Project Proposal	14
a. Background of the Study	14
b. Statement of the Problem	14
c. Objectives of the Study	14
d. Scope of the Study.....	14
e. Significance of the Study.....	14
f. Organisation of the Work.....	15
g. Time Management (Work Plan).....	15
h. Tools and Requirements.....	15

AIMS OF THE PROJECT

The project aims to develop students' research, analytical, and programming skills and to apply them to a substantial research project involving a well-defined scientific or engineering research or application problem chosen by the students (with guidance). The project will improve students' ability to utilise the knowledge, understanding, and skills acquired throughout the programme, integrating this in the context of the specific study and project undertaken.

INTENDED LEARNING OUTCOMES

Assessable outcomes

- ✓ The acquisition of detailed knowledge and understanding of a specialist subject
- ✓ The ability to source and reference material
- ✓ The ability to assess and critically appraise alternatives.
- ✓ The application of cognitive skills, including analysis and argument
- ✓ The application of problem-solving skills to particular research/scientific problems in the subject area
- ✓ The ability to design and implement an algorithm, to port an existing implementation to a new environment, to produce, test, and evaluate new or revised code.
- ✓ The application of written presentation skills and the formulation of results to be able to present defensible conclusions.
- ✓ The application of oral presentation skills and the ability to produce and use effective visual representations of research work.

Additional outcomes

- ✓ The acquisition of project-management skills
- ✓ To source information and employ information technology
- ✓ The ability to produce individual work

PROJECT SUPERVISION ACQUISITION

The student will, with appropriate guidance from the HOD or other lecturers, select a topic of interest to them and of academic/professional relevance within the area of research in the community. All students will be allocated a project supervisor who will act as an academic guide and provide support to the student. Each student will undertake their project research in one academic year and complete a written report and an oral presentation of their project by the official submission deadline.

BRIEF DESCRIPTION OF TEACHING AND LEARNING METHODS

Students will undertake their project under the guidance of their project supervisor(s). Interaction with the project supervisor(s) will be by means of individual tutorial support. Students are expected to be in touch with their supervisor(s) on a regular basis. A complete academic year will be devoted to the completion of the research and implementation stage, writing-up stage and the oral presentation of the dissertation.

ACADEMIC MISCONDUCT

Copying

It is clear to all that, if a student is asked to do a piece of work, then simply copying all or part of someone else's work is not acceptable: copying is cheating and regarded seriously as academic misconduct. More importantly, perhaps, it is not sensible. If there is a purpose in asking a student to do a piece of work, it will be to promote or assess the student's learning, and copying contributes nothing to that purpose. **Equally, you should never employ anyone to write an essay for you: this is a variant of copying.**

Cheating

Cheating is the attempt to gain an advantage for oneself or another by deceit and other misconduct, are breaches of discipline under the institution's regulations for conduct and is punishable by a range of sanctions.

Cheating includes:

- *Plagiarism*: Plagiarism is the fraudulent representation of another's work as one's own. This applies to whatever the source of the material (for example, a published source, the web, or the work of another student), whether the material is copied word for word or paraphrased, and whatever the extent of the material used.

Note:

All submitted project work must not exceed a similarity (plagiarism) index of 20%. Any work exceeding this threshold may be rejected or penalised

- *Use of AI-Generated Content*: The use of AI tools (e.g., ChatGPT and similar systems) must be **limited, transparent, and properly acknowledged**. Students are expected to demonstrate original understanding and independent work.

Note:

- AI-generated content in the project report **must not exceed 20%**.
- Students must be able to **explain and defend all parts of their work**, including any AI-assisted sections.
- Submitting AI-generated work as entirely original without disclosure constitutes academic misconduct.
- Falsifying data, evidence, or experimental results.
- Assisting another student in an act of plagiarism or falsification.

FORMATTING YOUR PROJECT

A well-presented dissertation enhances readability and reflects the quality of your work. Students are therefore required to adhere strictly to the following formatting guidelines:

- **Font Style**: Use *Times New Roman* throughout the document.
- **Font Size**: 12-point for all main text.
- **Line Spacing**: 1.5-line spacing must be used consistently.
- **Major Headings**: Should be **centred, bold, and clearly distinguishable** from the main text.

- **Minor Headings and Sub-headings:**
Should be **left-aligned, bold, and properly numbered** (e.g., 1.1, 2.1.1).
Capitalise major words in headings.
Numbering should **not exceed three levels** (maximum: 1.1.1). Avoid deeper levels such as 1.1.1.1.
- **Chapter Layout:**
Each chapter must **begin on a new page**.
- **Paragraph Spacing:**
Use consistent paragraph separation throughout the document.
You may either:
 - leave one blank line between paragraphs, or
 - apply spacing after paragraphs
(*choose one method and use it consistently*).
- **Text Alignment:**
All body text should be **justified** to ensure a clean and professional appearance.
- **Consistency:**
Maintain **uniform formatting for figures, tables, fonts, and graphical elements** throughout the entire document.

STRUCTURE OF THE REPORT

This section outlines the **required arrangement of the project report**. Students must follow this structure to ensure clarity, consistency, and professional presentation of their work.

1. **TITLE PAGE**, which should have the following format

KUMASI TECHNICAL UNIVERSITY
FACULTY OF APPLIED SCIENCES AND TECHNOLOGY
COMPUTER SCIENCE DEPARTMENT

<<Topic>>

**A PROJECT SUBMITTED TO THE COMPUTER SCIENCE DEPARTMENT
IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD
OF <<YOUR PROGRAMME OF STUDY>>**

<<Your name (First name, middle name and Surname)>>
(<<Student number>>)

<<MONTH, YEAR>>

2. DECLARATION (on a separate page)

The declaration is a formal statement by the student confirming that the project work is original and has not been submitted elsewhere for academic credit.

It should:

- State that the work is the student's own effort
- Confirm that all sources have been properly acknowledged
- Indicate that the work has not been submitted for another qualification

Note: The declaration must be signed and dated by the student.

3. CERTIFICATION (on the same page as Declaration)

The certification is an official statement by the project supervisor (and/or Head of Department) confirming that the work has been reviewed and approved.

It should:

- Confirm that the project has been supervised
- Indicate that the work meets the required academic standards
- Approving the project for submission

Note: The certification must be signed and dated by the supervisor and the HoD.

4. ACKNOWLEDGEMENTS (on a separate page)

This section allows the student to formally recognise individuals, groups, or institutions that contributed to the success of the project.

This may include supervisors, lecturers, family, sponsors, or organisations that provided support, guidance, or resources.

5. ABSTRACT (on a separate page)

The abstract provides a **concise summary of the entire project**. It is usually written last but placed at the beginning of the report.

The abstract should **not exceed 300 words** and must clearly present:

- Background of the study
- Problem statement
- Aim of the project
- Methodology used
- Tools/technologies used
- Key results
- Conclusion

Note: The abstract should be clear, brief, and self-contained, allowing a reader to understand the essence of the project without reading the full report.

6. TABLE OF CONTENTS (on separate page(s))

This section lists all chapters, sections, and subsections with their corresponding page numbers.

- Use **Roman numerals** (i, ii, iii, ...) for preliminary pages (Declaration, Acknowledgements, Abstract, Table of Contents, etc.).
- Use **Arabic numerals** (1, 2, 3, ...) from Chapter One onward.

- This numbering format also applies to:
 - List of Figures
 - List of Tables

7. LIST OF FIGURES (on a separate page)

Provide a complete list of all figures used in the report (excluding appendices), starting from Chapter One or Two as applicable.

Format:

Figure Number Figure Description Page Number

Example:

2.5 Architecture of Java SE 8

8. LIST OF TABLES (on a separate page)

Provide a comprehensive list of all tables used in the report (excluding appendices).

Format:

Table Number Table Description Page Number

Example:

3.1 Cost of Networking Materials in 2013 23

9. CONTENTS OF CHAPTERS (on separate pages)

Add the content of chapters from one to five. Each chapter should start on a fresh page

10. REFERENCES (on a separate page)

This section contains all sources cited in the report.

- Ensure all references are **accurately cited within the text**.
- Students must ensure that all sources cited in the text are properly listed in the reference section using the **APA (7th edition) style**.
- Include a variety of sources such as:
 - Books
 - Journals
 - Conference papers
 - Credible websites

Book

Format: Author(s). (Year). *Title of the book*. Publisher.

Example: Opoku, S. K. (2012). *Introduction to cryptography*. TechPress Publishing.

Journal Article

Format: Author(s). (Year). Title of the article. *Journal Name*, Volume(Issue), page range.
<https://doi.org/xxxxx>

Example: Mensah, K., & Boateng, A. (2020). Secure data transmission in distributed systems. *International Journal of Computer Science*, 15(2), 45–60.
<https://doi.org/10.1234/ijcs.2020.5678>

Conference Paper

Format: Author(s). (Year). Title of the paper. In *Name of Conference* (pp. xx–xx). Publisher.
DOI (if available)

Example: Asare, P., & Owusu, T. (2019). Machine learning approaches for fraud detection. In *Proceedings of the International Conference on Emerging Technologies* (pp. 120–125). IEEE.

Website

Format: Author(s) or Organization. (Year). *Title of the webpage*. URL

Example: World Health Organization. (2021). *Digital health technologies*.
<https://www.who.int/digital-health>

Online Article (No DOI)

Format: Author(s). (Year). Title of article. *Website Name*. URL

Example: Amoako, J. (2022). Cybersecurity trends in Africa. *Tech Africa News*.
<https://www.techafrika.com/cybersecurity-trends>

Thesis / Dissertation

Format: Author. (Year). *Title of thesis* (Degree type, Institution). Repository/URL

Example: Adjei, M. (2021). *Design of a secure e-voting system* (Master's thesis, Kumasi Technical University).

11. APPENDICES (on a separate page)

The appendices contain **supporting materials** that are too detailed to include in the main body of the report.

Examples include:

- Questionnaire or survey instruments
- Interview questions
- Source code
- System screenshots
- Test cases and results
- User manual
- Gantt chart
- Budget
- Additional diagrams

PROJECT OUTLINE / CONTENTS OF CHAPTERS

This section guides the **expected content of each chapter**. Students should follow these guidelines, although minor adjustments may be made in consultation with their supervisor.

Chapter One: Introduction

This chapter introduces the project and explains its importance.

It should include:

- Background of the Study
- Problem Statement
- Aim of the Study
- Objectives of the Study
- Research Questions (optional)
- Significance of the Study
- Scope of the Study
- Limitations of the Study
- Organisation of the Report

Chapter Two: Literature Review

This chapter examines existing research and related work.

It should include:

- Conceptual review of key terms and ideas
- Review of existing systems
- Review of related technologies
- Review of methods, algorithms, or models
- Identification of gaps in existing systems
- Summary of the literature

All sources must be properly cited.

Chapter Three: Methodology and System Design

This chapter explains how the project was carried out and how the system was designed.

It should include:

Research Methodology

- Research approach (e.g., Design Science, Survey, Experimental)
- Data collection methods (questionnaire, interview, dataset)
- System development methodology (Agile, Waterfall, Scrum, RAD)

Requirements Analysis

- Functional requirements
- Non-functional requirements
- User requirements / user stories

Tools and Technologies

- Programming languages
- Frameworks/libraries
- Database systems
- Development tools (e.g., IDEs)

System Design

- System architecture diagram

- Use case diagram
- Database design (ERD)
- Add 2 – 3 of the following:
 -  Activity diagram
 -  Sequence diagram
 -  Class diagram
 -  Input design
 -  Output design

Algorithm / Model Description

- Pseudocode, flowcharts, or algorithm explanation

Data Description (if applicable)

- Source of data
- Size and format
- Preprocessing steps

Validation / Testing Plan

- Testing approach
- Types of testing (unit, integration, user testing)

Ethical Considerations

- Data privacy
- Consent (if human data is used)
- Security considerations

Chapter Four: System Implementation, Testing and Results

This chapter presents how the system was developed and evaluated.

It should include:

- Development environment
- Programming language(s) used
- Tools and frameworks
- Database implementation
- System modules/components
- Interface design (with screenshots)
- Explanation of key code modules
- Test plan and test cases
- Test results (tables)
- System performance results
- Evaluation of the system

Chapter Five: Discussion, Conclusion and Recommendations

This chapter interprets results and highlights contributions.

It should include:

- Summary of findings
- Interpretation of results
- Whether objectives were achieved
- Comparison with existing systems
- Advantages of the new system
- Challenges encountered

- Implications of the study
- Contribution of the project
- Recommendations for future work

Final Submission Requirements

- All projects must include:
 - **Similarity (plagiarism) report**
 - **AI-generated content report**
- The following limits must be strictly observed:
 - **Similarity index $\leq 20\%$**
 - **AI-generated content $\leq 20\%$**
- Projects exceeding these thresholds may be penalised or rejected.

ASSESSMENT

Criteria	Description	Maximum Mark
Report (60%)	Abstract	5
	Introduction	5
	Literature Review	10
	Methodology and System Design	10
	System Implementation, Testing and Results	15
	Discussion, Conclusion and Recommendations	10
	References	5
Defence (40%)	Jury Award for the Artefact: This assesses the quality, functionality, and innovation of the developed system	40
	Total	100%

WRITING YOUR PROJECT PROPOSAL

A good proposal is likely to be used as chapter one of your project when written properly. You are therefore advised to take your time and think deeply about what you want to do.

Title Page Format

KUMASI TECHNICAL UNIVERSITY
FACULTY OF APPLIED SCIENCES AND TECHNOLOGY
COMPUTER SCIENCE DEPARTMENT

<<Topic>>

PROJECT PROPOSAL

<<Your name (First name, middle name and Surname)>>

(<<Student number>>)

<<Programme of Study>>

SUPERVISOR

<<Supervisor's name (First name, middle name and Surname)>>

<<Date>>

Content of the Project Proposal

This section outlines the key components that must be included in a project proposal. Students should present each subsection clearly and concisely.

a. Background of the Study

This section provides the context and overview of the research topic. Students should:

- Introduce the general area of study
- Explain the context and relevance of the topic
- Provide supporting information from existing studies (where necessary)
- Lead logically to the problem being addressed

The background should be clear, focused, and informative, helping the reader understand why the study is important.

b. Statement of the Problem

This is the core issue the project seeks to address. Students should:

- Clearly state the problem in one concise sentence
- Provide supporting explanation
- Use evidence or examples to show the seriousness of the problem
- State why the need to solve the problem

A good problem statement is specific, measurable, and researchable.

c. Objectives of the Study

This section outlines what the project intends to achieve. Students should include:

Aim (overall aim of the project)

Specific Objectives (3–5 clear and measurable goals)

Objectives should be action-oriented (e.g., design, develop, evaluate, implement).

d. Scope of the Study

This defines the boundaries and limitations of the project. Students should specify:

- What the project will cover
- What will not be covered
- Target users or population (if applicable)
- Location or environment of the study
- Duration and constraints

This helps manage expectations and keeps the project focused.

e. Significance of the Study

This explains the importance and benefits of the project. Students should:

- Identify who will benefit (e.g., users, organizations, researchers)

- Explain how the project contributes to knowledge or practice
- Highlight practical or societal impact

f. Organisation of the Work

This provides a brief overview of the structure of the report. Students should summarise each chapter in 1–2 sentences:

- Chapter One: Introduction
- Chapter Two: Literature Review
- Chapter Three: Methodology and System Design
- Chapter Four: System Implementation, Testing and Results
- Chapter Five: Discussion, Conclusion and Recommendations

g. Time Management (Work Plan)

This section outlines how the project will be completed within the given timeframe. Students should:

- Present a timeline (preferably using a Gantt chart)
- Show key phases such as:
 - Proposal
 - Literature review
 - Design
 - Implementation
 - Testing
 - Report writing

The schedule should be realistic and achievable.

h. Tools and Requirements

This section describes the resources needed for the project.

- For software projects:
 - Programming language(s)
 - Frameworks/libraries
 - Database systems
 - Development tools (e.g., IDEs)
- For non-software projects:
 - Equipment and materials
 - Software tools
 - Data sources
 - Method of acquisition

Students should justify why the selected tools are appropriate.